

EC-201: Engineering Project Management

Project Life cycles

Broad Contents

- Life cycle phases of a Product
- Life cycle phases of a System
- System Costs
- Cost Benefit Analysis
- Explanation of Various Life Cycle Phases
- Characteristics of Project Life Cycle
- Project Management Office
- Project Management Officer (PMO)
- Difference between Project Manager and Project Management Officer
- Some Examples of Project Life Cycle

5.1 Life Cycle Phases of a Product

- Every program, project, or product has certain phases of development. A clear understanding of these phases permits managers and executives to better control total resources in the achievement of desired goals. The phases of development are known as life-cycle phases. However, the breakdown and terminology of these phases differ, depending on whether we are discussing products or projects.
- During the past few years, there has been at least partial agreement about the life cycle phases of a product. They include:
 - Research and development
 - Market introduction
 - Growth
 - Maturity
 - Deterioration
 - Death

5.2 Life Cycle Phases of a System (1/8)

- The theoretical definitions of the life cycle phases of a system can be applied to a project, as shown in Fig 5.1. These phases include:
 - Conceptual
 - Planning
 - Testing
 - Implementation
 - Closure
- **PMO** stands for the present method of operations, and **PMO'** will be the "new" present method of operations after conversion.

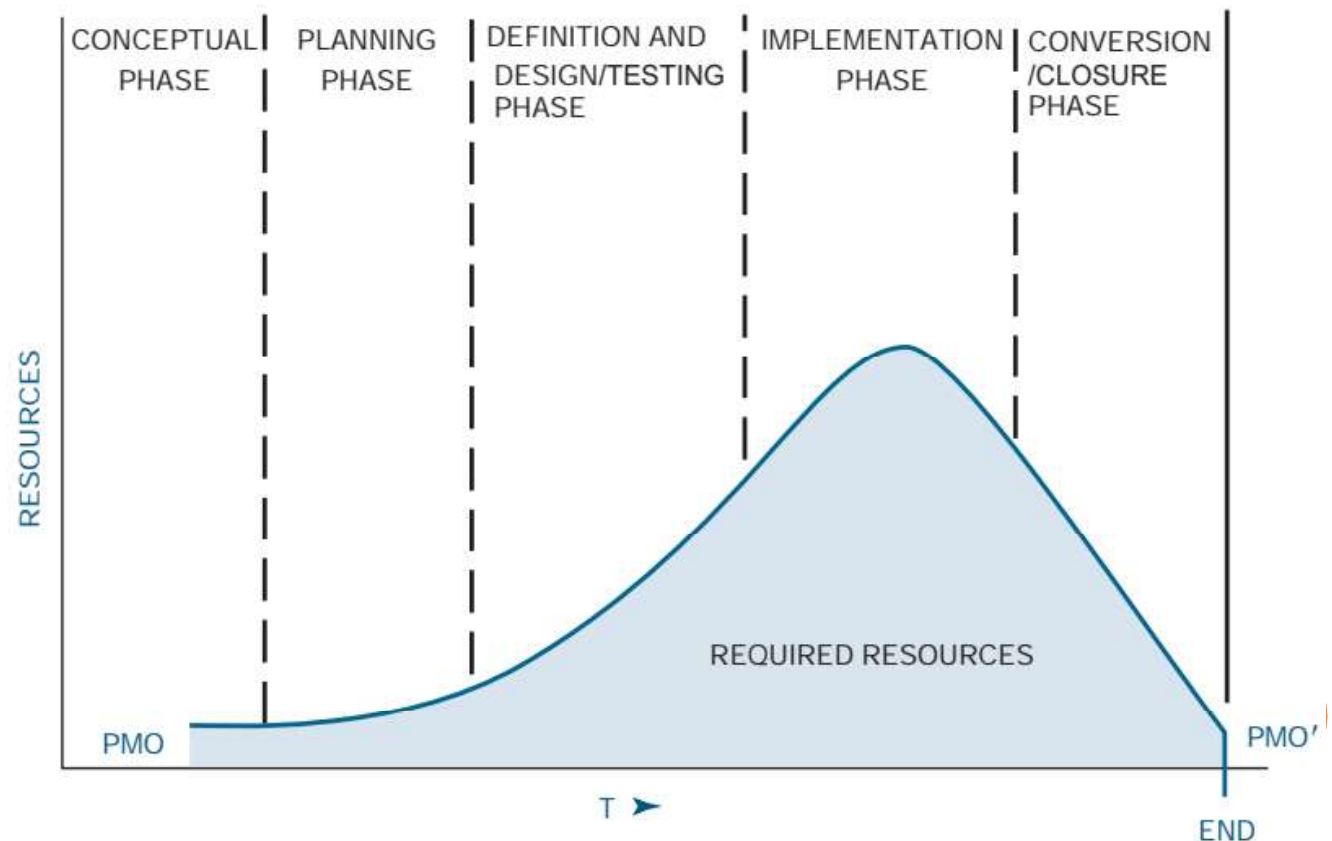


Fig 5.1: Definition of a Project Life Cycle

5.2 Life Cycle Phases of a System (2/8)

- a. *Conceptual Phase:*** The first phase, the conceptual phase, includes the prior evaluation of an idea. Most important in this phase is a prior analysis of risk and the resulting impact on the time, cost, and performance requirements, together with the potential impact on company resources.
- b. *Planning Phase:*** The second phase is the planning phase. It is mainly a refinement of the elements described under the conceptual phase. The planning phase requires a firm identification of the resources to be required together with the establishment of realistic time, cost, and performance parameters. This phase also includes the initial preparation of all documentation necessary to support the system.
 - i. *System Costs:*** Because of the amount of estimating involved, analysing system costs during the conceptual and planning phases is not an easy task.

5.2 Life Cycle Phases of a System (3/8)

b. Planning Phase:

- i. System Costs (continued...):* As shown in Figure 5.2 next slide, most project or system costs can be broken down into operating (recurring) and implementation (nonrecurring) categories. The implementation costs include one-time expenses such as construction of a new facility, purchasing computer hardware etc. Operating costs, on the other hand, include recurring expenses such as manpower. The operating costs may be reduced due to improved position on learning curve. The learning curve is a graphical representation of repetitive tasks that, when done on a continuous basis, lead to a reduction on activity duration, resources and costs.

5.2 Life Cycle Phases of a System (4/8)

b. Planning Phase:

i. System Costs:

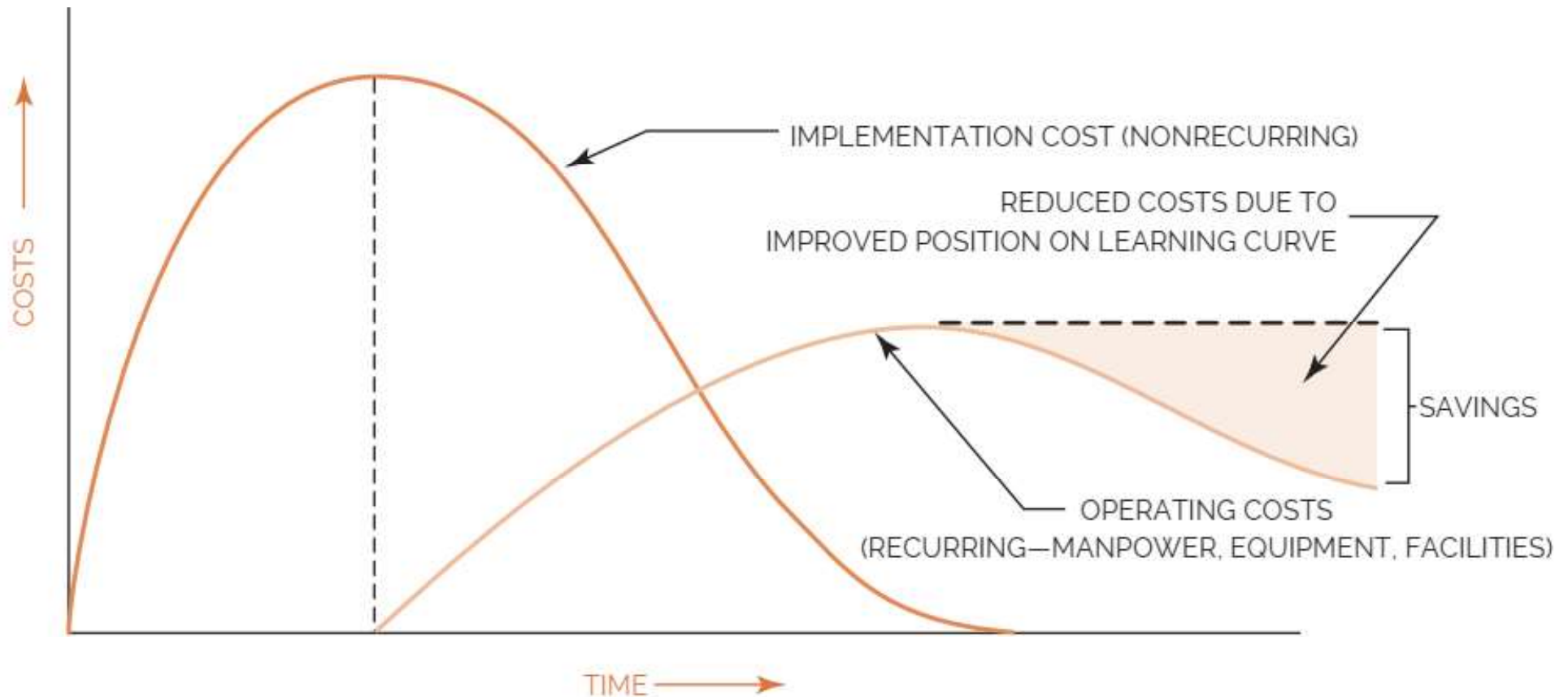


Fig 5.2: System Costs

5.2 Life Cycle Phases of a System (5/8)

b. Planning Phase:

ii. Cost Benefit Analysis (CBA): Once the approximate total cost of the project is determined, a cost-benefit analysis should be conducted (see Figure 5.3 next slide) to determine if the estimated value of information (VOI) obtained from the system exceeds the cost of obtaining the information. This analysis is often included as part of a feasibility study. Value of Information (VOI) is the amount a decision maker would be willing to pay for information prior to making a decision.

c. Testing Phase: The third phase — testing — is predominantly a testing and final standardization effort so that operations can begin. Almost all documentation must be completed in this phase.

5.2 Life Cycle Phases of a System (6/8)

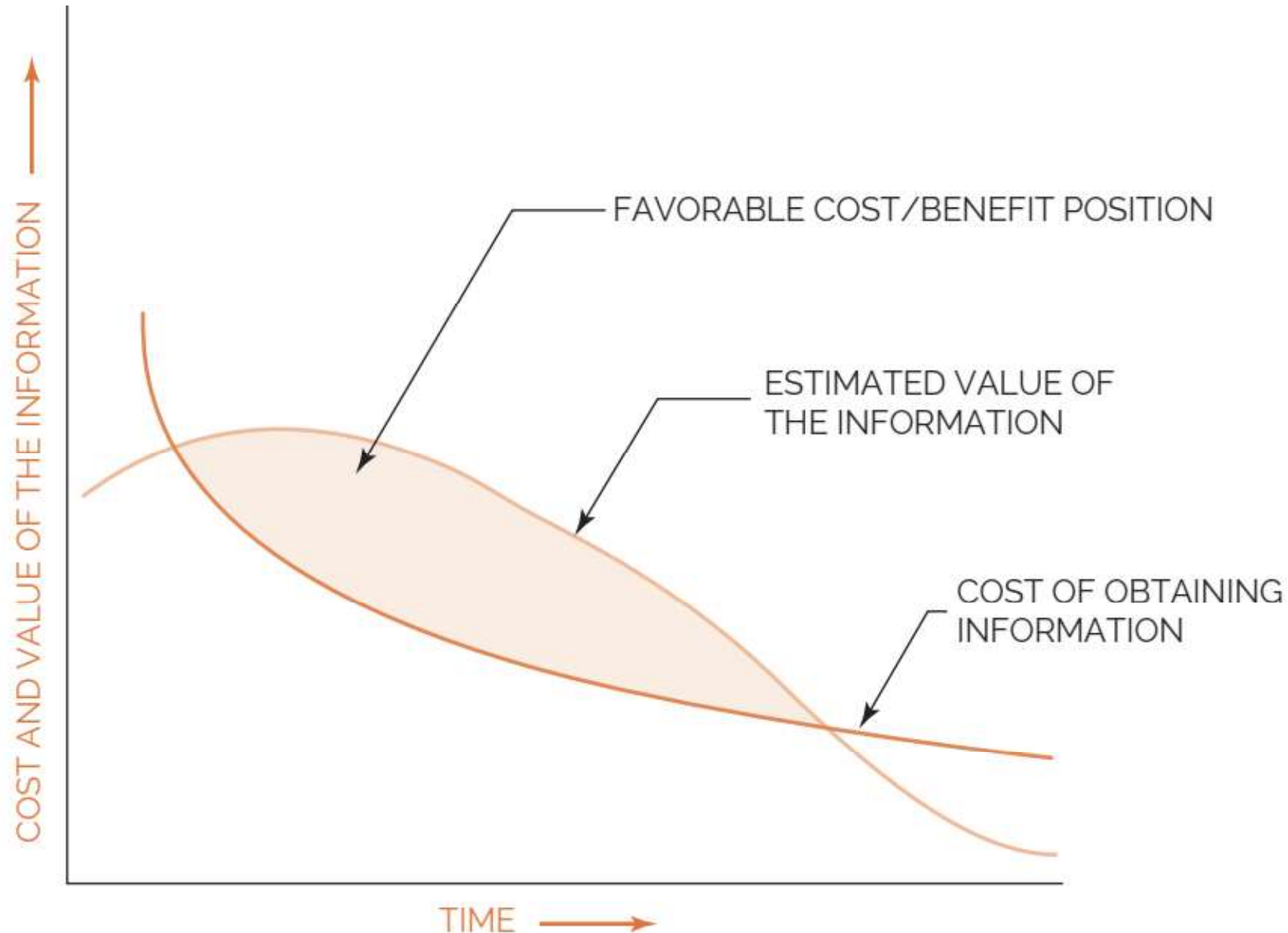


Fig 5.3: Cost-Benefit-Analysis

5.2 Life Cycle Phases of a System (7/8)

- d. Implementation Phase:* The fourth phase is the implementation phase, which integrates the project's product or services into the existing organization. If the project was developed for establishment of a marketable product, then this phase could include the product life cycle phases of market introduction, growth, maturity, and a portion of deterioration.
- e. Closure Phase:* The final phase is closure and includes the reallocation of resources. The question to be answered is, "Where the resources should be reassigned?" Consider a company that sells products on the open consumer market. As one product begins the deterioration and death phases of its life cycle, then new products or projects must be established. Such a company would, therefore, require a continuous stream of projects as a necessity for survival, as shown in Figure 5.4 next slide. As projects A and B begin their decline, new efforts (project C) must be developed for resource reallocation.

5.2 Life Cycle Phases of a System (8/8)

- e. **Closure Phase (continued...):** The closure phase evaluates the efforts on the total system and serves as input to the conceptual phases for new projects and systems.

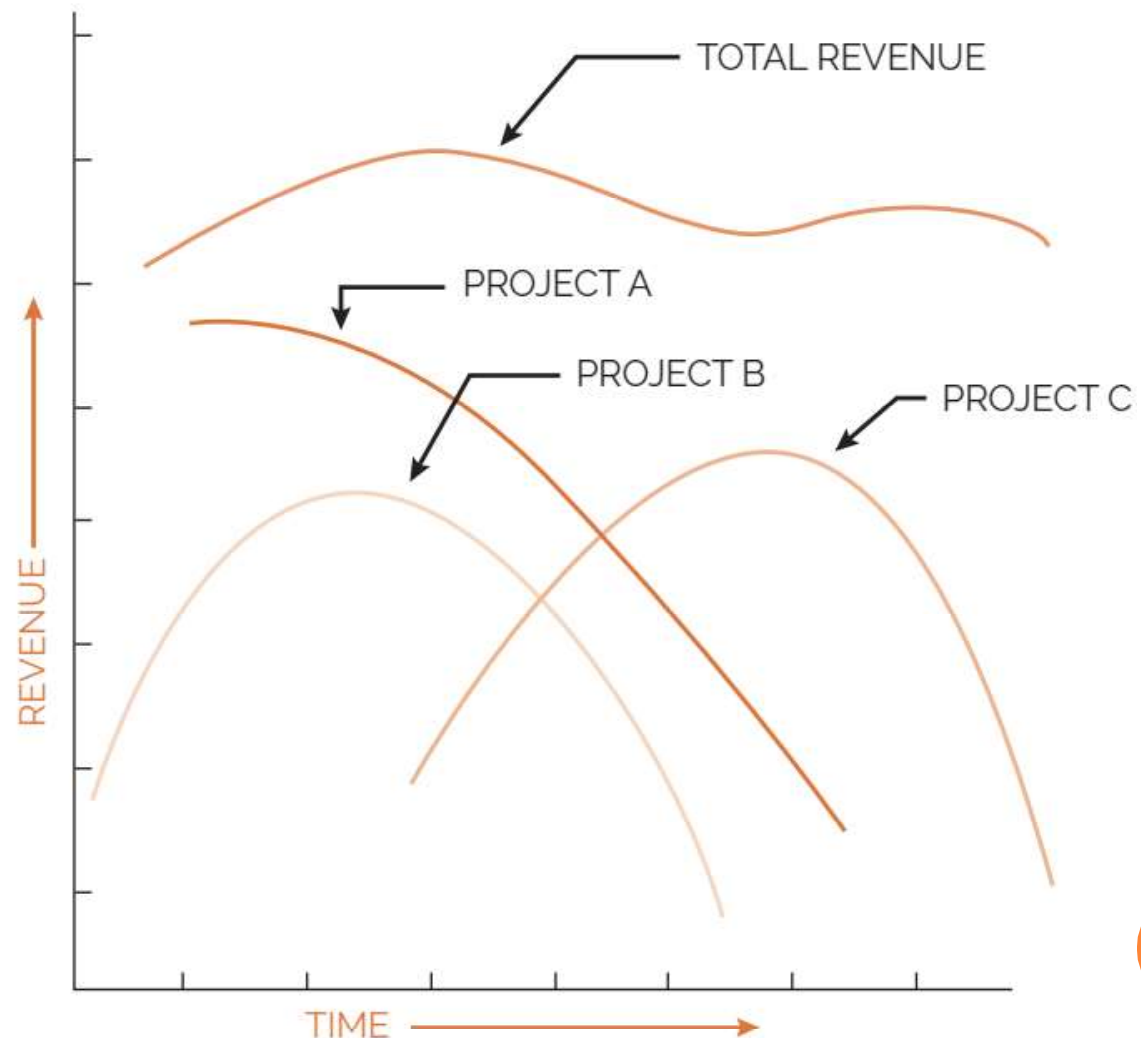


Fig 5.4: A Stream of Projects

5.3 Explanation of Various Life Cycle Phases

- Table 5.1 identifies the various life cycle phases that are commonly used. Even in mature project management industries such as construction, one could survey ten different construction companies and find ten different definitions for the life cycle phases.

TABLE 5-1 LIFE-CYCLE PHASE DEFINITIONS

Engineering	Manufacturing	Computer Programming	Construction
• Start-up • Definition • Main • Termination	• Formation • Buildup • Production • Phase-out • Final audit	• Conceptual • Planning • Definition and design • Implementation • Conversion	• Planning, data gathering, and procedures • Studies and basic engineering • Major review • Detail engineering • Detail engineering/construction overlap • Construction • Testing and commissioning

5.4 Characteristics of Project Life Cycle (1 / 2)

- Project life cycle defines phases that connect beginning and end of the project. After each phase deliverables are reviewed for the completeness in time, accuracy according to defined objectives.
- Project life cycle phases can be overlapped to save time and to have fast tracking on the life cycle.
- There is no way to define Project Life Cycle ideally. Only three phases are always certain to be performed; conceptualization, intermediate phase(s), and closure.
- Generally phases are defined in sequential order by technical information officer.
- Cost and staffing level is defined for every single phase.
- Project may have sub-project(s) and sub-projects may have their own project life cycle.
- In the beginning of the project, level of uncertainty and risk is always high.

5.4 Characteristics of Project Life Cycle (2/2)

- The typical project life cycle – initiating, implementing and closing – has critical decision points where the project may continue, be changed, or be abandoned.
- There are many points within the project life cycle where *Community of Professionals (COPs)* may provide support and guidance.
- During project close, reassignment and intelligent preservation of resources, deliverables, and sharing lessons learned are facilitated.

5.5 Project Management Office (1/2)

The Project Management Office sets project standards and oversees the organization's portfolio of projects. This allows the organization to evaluate the use of resources across all projects and resolve conflicts that affect project timelines. The Project Management Office is also a very good place to examine how communities are linked across projects. Using the communities as the linkage point for knowledge transfer is far more efficient for the following several reasons:

- In communities, the evaluation of knowledge is generally done by a broader range of people, ensuring that the ideas are more completely vetted.
- Communities generally exist outside the project framework and trust is already established.
- In communities, knowledge is transferred from expert to recipient. Generally knowledge is transferred via documents.

5.5 Project Management Office (2/2)

- Community shares the knowledge broadly, strengthening the entire organization for future projects.
- Project Management Office and top management are responsible for the periodic review of major projects.

5.6 Project Management Officer [PMO] (1/2)

Project Management Officer (PMO) centralizes and coordinates management of project under his domain, and oversees management of project and product (system/program) both. He/she focuses on coordination planning, prioritization of all resources and deliverables of projects and subprojects. It is the responsibility of Project Management Officer to keep top management and clients/parents organization connected and informed about all projects running or product lifecycle. Project Management Officer is generally responsible for:

- Providing monitoring platform for Project Manager.
- Identifying the Project Management Methodology and best practices for specific project.
- Clearing house, i.e. defining and refining project policies, procedures, templates and shared documents.
- Configuration management for all projects under work.

5.6 Project Management Officer [PMO] (2/2)

- For developing and keeping repository and risk management for projects.
- Developing Project Management Office for operation and maintaining tools for project management.
- Management, coordination and monitoring of communications across the projects, project timelines and budget, quality standards.
- Project Management Officer may be having authority to terminate project anytime when he gets it not feasible anymore.

5.7 Difference Between Project Manager and Project Management Officer: (1 / 2)

1. Both have different objectives- driven by different requirements, aligned with strategic needs of organization.
2. Project Manager is responsible for delivering specific project objectives within project constraints, while Project Management Officer is responsible for organizational structure specific mandates having much vast perspective.
3. Project Manager focuses on project objectives, while Project Management Officer focuses on major programs, scope and changes required and authenticated.
4. Project Manager is constrained with assigned resources for specific project to meet its full objective. On the other hand, Project Management Officer is supposed to optimize the use of shared organizational resources across all projects overall.

5.7 Difference Between Project Manager and Project Management Officer: (2/2)

5. Project Manager manages scope, schedule, cost, and quality of product, while the Project Management Officer manages overall risk, opportunities, interdependencies and links among different projects.
6. Project Manager reports on project progress/project specific information to the top management, while Project Management Officer provides consolidated reporting on all the running projects.

5.8 Some Examples of Project Life Cycle (1/3)

The project life cycle is conveniently represented by a bar chart which clearly indicates the duration of each phase and its overlap (if any) with the other phases.

- i. **House Project:** The construction of a house provides a project many of us have personally experienced. Consider here the following simple subdivision into four phases.

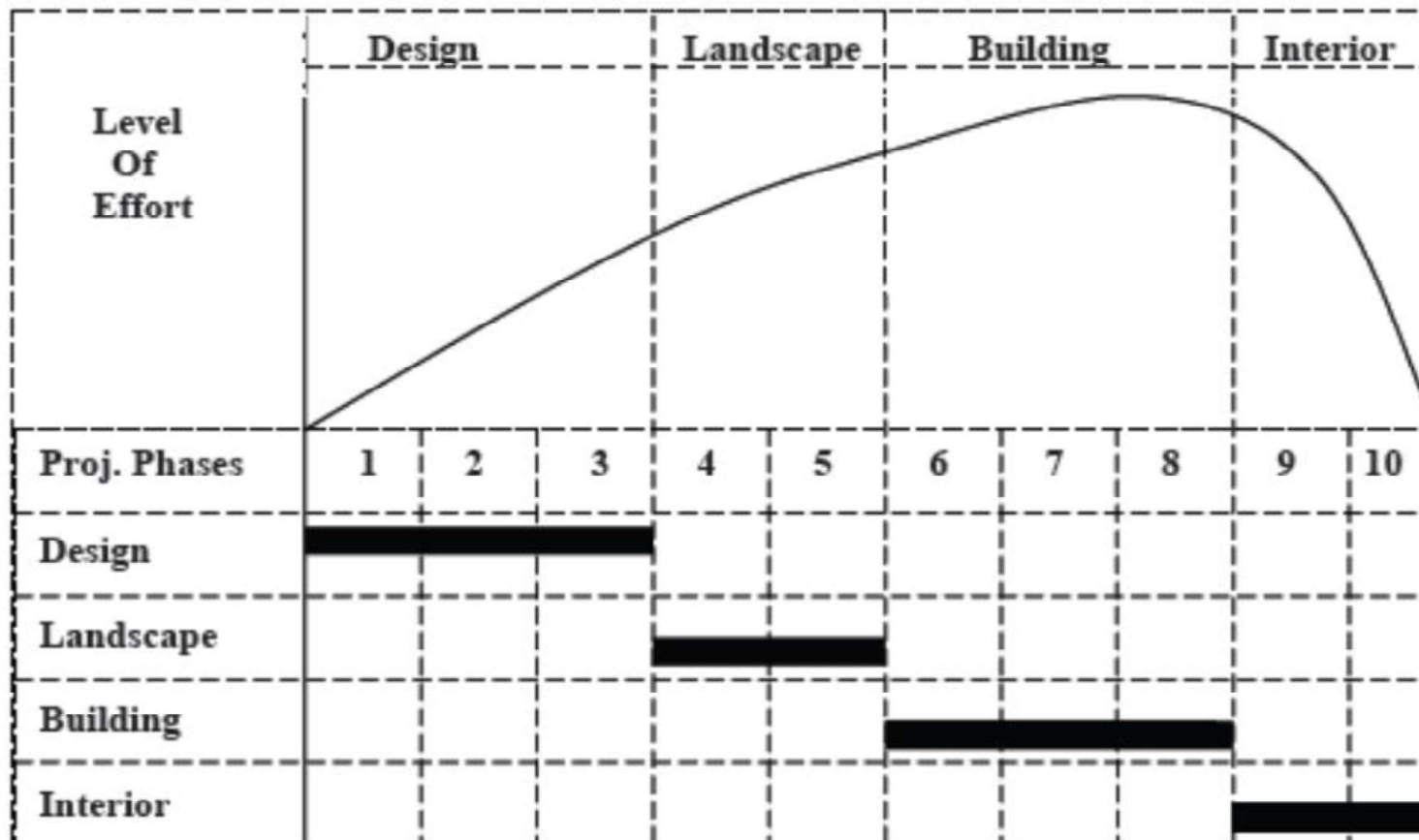


Fig 5.5: Project Life Cycle: House Project

5.8 Some Examples of Project Life Cycle (2/3)

- ii. **Computer installation:** With the improved cost effectiveness of computer facilities most companies will experience a computer installation project sooner or later.

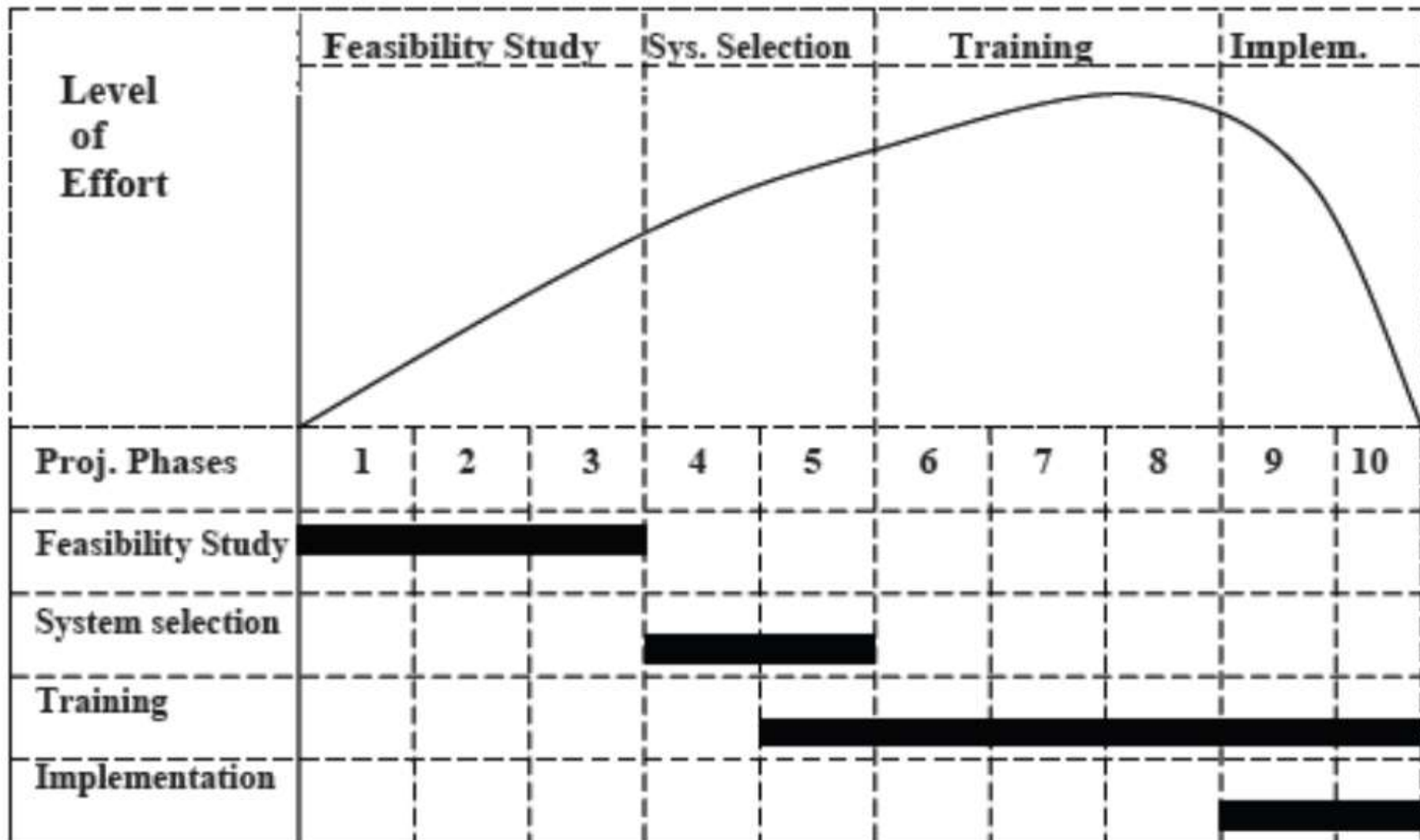


Figure 5.6: Project Life Cycle: Computer Installation

5.8 Some Examples of Project Life Cycle (3/3)

- iii. Engineering Project:* An engineering type project is a popular example to illustrate the project phases.

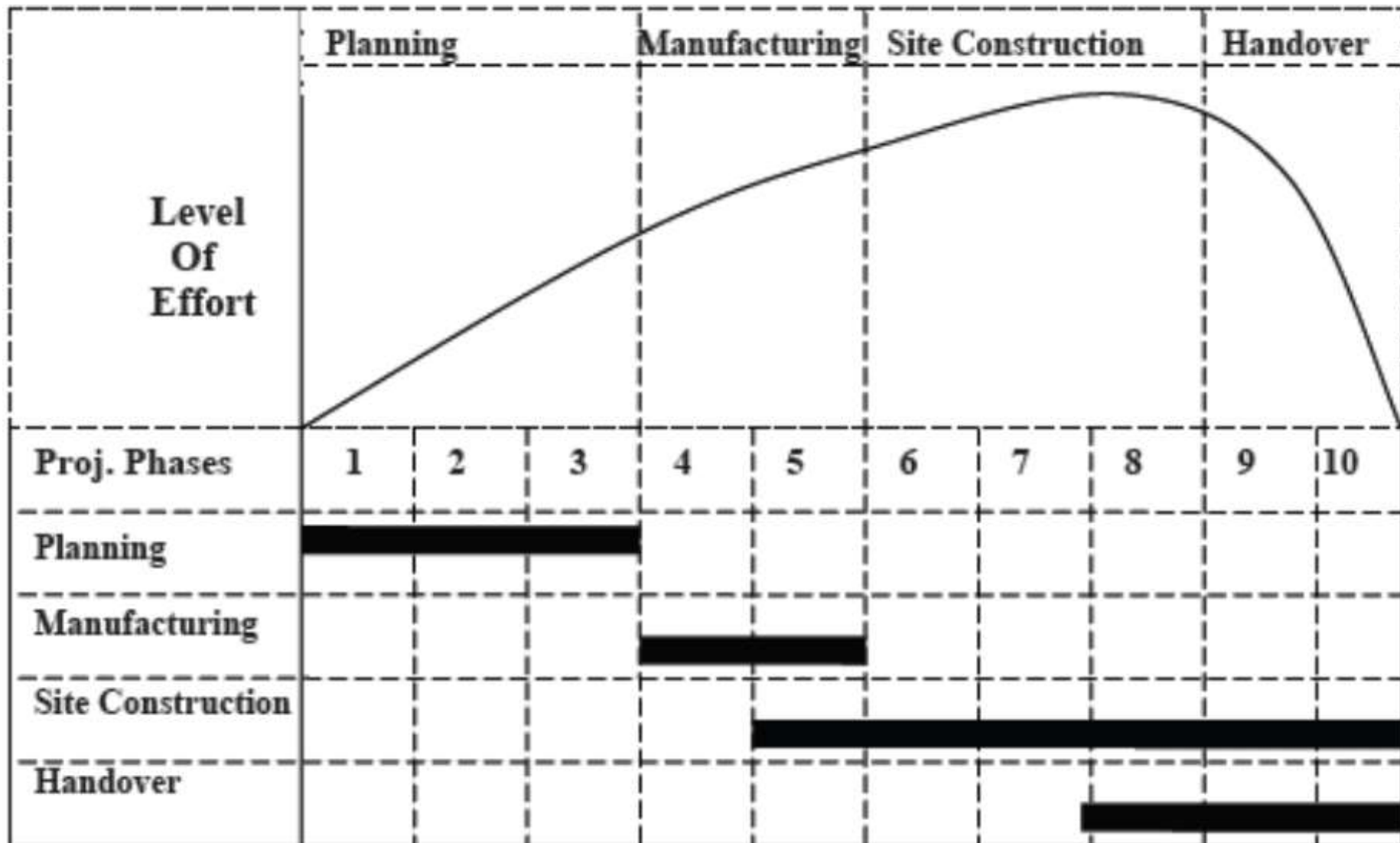


Figure 5.7: Project Life Cycle: Engineering Project

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† Timothy J. Kloppenborg, Vittal Anantatmula, Kathryn N. Wells, *“Contemporary Project Management”* 4th Edition, Cengage Learning.

